AYMANE AIT DADS

Software Development Intern | AI-Enhanced Engineering · Machine Learning Pipelines · End-to-End Deployment

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PROFESSIONAL SUMMARY

End-to-end ML pipeline engineer who built a Random Forest and K-Means classification system on 100K+ fiber-optic network records at Orange Maroc, reducing manual analysis time by 60% and delivering direct output to the network optimization team. Ranked top 10% on the NVIDIA Nemotron Reasoning Kaggle leaderboard (0.80/1.0, $106K+ prize pool) by fine-tuning a 30B Mamba-Transformer MoE model with LoRA on a Blackwell GPU, resolving 5 architecture-level incompatibilities through systematic ablation studies. Core competencies span Python, PyTorch, Hugging Face Transformers, feature engineering, clustering, REST APIs, and AI-enhanced requirements engineering workflows. Available for internship immediately; no South America work authorization required as a French-based student eligible for international internship programs.

CORE COMPETENCIES

AI-Enhanced Engineering | Machine Learning Pipelines | Requirements Engineering | Feature Engineering | Stakeholder Communication | Software Development | Data-Driven Optimization | Ablation Studies

WORK EXPERIENCE

Orange Maroc
Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built an end-to-end machine learning pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Engineered a clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production maintenance decisions.
- Automated feature engineering and cross-validated model evaluation workflows, reducing manual analysis time by **~60%** and accelerating reporting cycles for high-level stakeholders.
- Delivered stakeholder presentations translating model outputs into actionable maintenance recommendations, producing **Power BI dashboards** that communicated data-driven findings to non-technical program managers.

PROJECTS

NVIDIA Nemotron Reasoning Challenge
Kaggle Competition · In Progress (June 2026) · Top 10% Public Leaderboard

- Achieved **top 10% public leaderboard (0.80/1.0)** in the $106K+ prize pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA (r=32, ~888M trainable parameters) on 7,828 chain-of-thought logic reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequential order - including OOM monkey-patching, Mamba CUDA kernel disabling, and Triton ptxas fallbacks - and conducted 20+ single-variable ablations confirming that packing degraded score from 0.80 to 0.64 and synthetic data consistently underperformed.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

AI-Generated Image Detection
EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st on the EURECOM private Kaggle class leaderboard (0.791 AUC)** by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning rates across 250K training images spanning 25+ AI generator types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Identified that **1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC)** via systematic ablation on out-of-distribution generalization, then applied a 3-model ensemble weighted by validation AUC with 10-view TTA to achieve the final leaderboard score.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM
Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya
2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | TRL | PEFT | Unsloth | TensorFlow

AI & LLM Development: LoRA | QLoRA | SFTTrainer | vLLM | RAG | LangChain | Prompt Engineering | Chain-of-Thought

Data Science & Infrastructure: Python | SQL | Pandas | NumPy | Power BI | Git | Docker | PostgreSQL | REST APIs